

Recommendations — osteosarcoma-mets-dll3-h7r2

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** bone
- **Histology:** osteosarcoma
- **Stage:** IV
- **ECOG:** 1
- **Age band:** 18-29
- **Sex:** unknown
- **Biomarkers:** DLL3 RNA expression elevated; protein-level membrane status unknown (rna_only); PRAME RNA expression elevated; protein-level expression and HLA presentation status unknown (rna_only)
- **Efficacy/toxicity weight:** 0.85

6 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

DLL3-targeting interventions

1. tarlatamab (DLL3 x CD3 BiTE) via NCT06788938 (DLL3-IHC-gated basket) *(conditional on dll3_ihc positive)*

Likelihood of effect: Moderate if DLL3 IHC confirms — best-validated mechanism (DeLLphi-304 OS HR 0.60) but efficacy is entirely SCLC-derived; no osteosarcoma data.

Toxicity burden: Moderate (CRS ~56% mostly G1-2, ICANS-like ~12%, G3+ TRAE 24%)

Counter-productive MoA: **Moderate** (On-mechanism CNS/neuro bystander activation and antigen-density-dependent escape if DLL3 membrane expression is low or heterogeneous)

Overall: The best-validated mechanism on the board and the lighter-toxicity DLL3 route — but every efficacy row is SCLC; osteosarcoma activity is a hypothesis until the IHC confirms surface target.

Key references: PMID 40454646 · PMID 37861218 · NCT06788938

2. tarlatamab + radiation therapy via NCT06814496 (pan-tumor DLL3-IHC basket with RT) *(conditional on dll3_ihc positive)*

Likelihood of effect: Unknown — no combination data; abscopal benefit over tarlatamab monotherapy is a hypothesis the phase-1 is still measuring.

Toxicity burden: Moderate (uncharacterized combination; BiTE CRS ~50% mostly G1-2, ICANS-like ~10%, plus site-specific RT toxicity)

Counter-productive MoA: **Moderate** (Uncharacterized RT-on-BiTE synergistic toxicity; potential on-mechanism

CNS/neuro bystander activation compounded by radiation)

Overall:A parallel DLL3 slot with an abscopal rationale — but zero combination data, and the monotherapy basket delivers the same backbone with a written safety algorithm.

Key references: NCT06814496 · PMID 40454646

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
SHR-4849 (IDE849) (basket/biomarker)	ADC	1/2	SCLC, neuroendocrine carcinomas, DLL3-expressing tumors	recruiting	NCT07174583
alveltamig	trispecific	3	relapsed SCLC	recruiting	NCT07189455
zocilurtatug pelitecan	ADC	3	small-cell lung cancer	recruiting	NCT07218146
obrixtamig	BiTE	3	extensive-stage SCLC (1L)	recruiting	NCT07472517
gocatamig	trispecific	1/2	DLL3-expressing SCLC, large-cell NEC, neuroendocrine prostate cancer, GEP-NEC	active not recruiting	NCT04471727
peluntamig	bispecific_other	1/2	SCLC, LCNEC, NEPC, GEP-NEC, EP-NEC	recruiting	NCT05652686
NG2 and DLL3 CAR-T cells	CAR-T	1/2	melanoma	recruiting	NCT07193966
[212Pb]Pb-MP0712 (212Pb-DOTAM-MAM279)	radioligand	1/2	SCLC and other DLL3-expressing solid tumors (LCNEC, EP-NEC, GEP-NEC, NEC of bladder)	recruiting	NCT07278479
DJI136	CAR-T	1/2	extensive-stage SCLC	recruiting	NCT07564401
QLS31904	BiTE	1	advanced solid tumors with DLL3 expression	unknown	NCT05461287
DLL3-CAR-NK cells	CAR-NK	1	extensive-stage SCLC	unknown	NCT05507593
clesitamig	trispecific	1	SCLC, neuroendocrine carcinoma	completed	NCT05619744
LB2102	CAR-T	1	extensive-stage SCLC, LCNEC of lung	active not recruiting	NCT05680922
FZ-AD005	ADC	1	advanced solid tumors, SCLC, LCNEC	recruiting	NCT06424665
IBI3009	ADC	1	small-cell lung cancer	recruiting	NCT06613009
²² Ac-ABD147	radioligand	1	SCLC, large-cell neuroendocrine carcinoma	active not recruiting	NCT06736418

¹ Lu-DTPA-SC16.56		1	neuroendocrine tumors / carcinomas (lung, prostate)	not yet recruiting	NCT06941480
²² Ac-ETN029	radioligand	1	SCLC, LCNEC, NEPC, GEP-NEC	recruiting	NCT07006727
BL-M14D1	ADC	1	SCLC and neuroendocrine neoplasms (NEPC, GEP-NEC, Merkel cell, LCNEC, EP-NEC)	recruiting	NCT07080242
NEUK203-13 (anti-DLL3 iPSC CAR-NK)	CAR-NK	1	extensive-stage SCLC and neuroendocrine tumors	recruiting	NCT07366658
ML261	CAR-T	1	R/R SCLC and select NECs (EP-NEC, GEP-NEC, NEPC)	not yet recruiting	NCT07488923
rovalpituzumab tesirine (discontinued)	ADC	3	small-cell lung cancer (2L)	completed	NCT03061812
SC-002 (discontinued)	ADC	1	small-cell lung cancer	terminated	NCT02500914
AMG 119 (discontinued)	CAR-T	1	small-cell lung cancer	unknown	NCT03392064

Evidence in detail

tarlatamab (DLL3 x CD3 BiTE)

****Mountzios et al. *N Engl J Med* 2025****

- **Disease context:**SCLC after platinum-based chemo (2L) — Platinum-resistant or platinum-sensitive SCLC; ECOG 0-1
- **n:** 509
- **Toxicities:**CRS (grade any): 56% (n=254); **CRS** (grade ≥3): 1% (n=254); **ICANS-like neurologic event** (grade any): 12% (n=254); **treatment-related AE** (grade ≥3): 24% (n=254); **neutropenia** (grade ≥3): 3% (n=254)
- **Efficacy:**HR_OS 0.6 HR (95% CI 0.47-0.77); **OS** 13.6 months; **OS** 8.3 months; **PFS** 4.2 months; **PFS** 3.2 months; **TRAE_rate** 24%

****Ahn et al. *N Engl J Med* 2023****

- **Disease context:**previously treated SCLC (2L+) — SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated
- **n:** 220
- **Toxicities:**CRS (grade any): 51%; **CRS** (grade ≥3): 1%; **ICANS-like neurologic event** (grade any): 10%; **treatment-related AE** (grade ≥3): 30%; **treatment discontinuation due to AE** (grade any): 3%
- **Efficacy:**ORR 40% (95% CI 29-52); **DoR** not reached months; **PFS** 4.9 months; **TRAE_rate** 30%

tarlatamab + radiation therapy

****Mountzios et al. *N Engl J Med* 2025****

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- **n:** 509

- **Toxicities:** CRS (grade any): 56% (n=254); CRS (grade ≥ 3): 1% (n=254); ICANS-like neurologic event (grade any): 12% (n=254); treatment-related AE (grade ≥ 3): 24% (n=254); neutropenia (grade ≥ 3): 3% (n=254)
- **Efficacy:** HR_OS 0.6 HR (95% CI 0.47-0.77); OS 13.6 months; OS 8.3 months; PFS 4.2 months; PFS 3.2 months; TRAE_rate 24%

Prame Hla interventions

1. IMA203 (PRAME TCR-T) via NCT03686124 (ACTengine pan-solid basket, sarcoma cohort named) *(conditional on prame_hla positive)*

Likelihood of effect: High ceiling if both biomarkers confirm (ORR 54%, Nat Med 2025) but from n=28 with a wide CI and no osteosarcoma-specific responder; unstable point estimate.

Toxicity burden: High (1 treatment-related septic death; CRS ~100%/G3+ 11%, ICANS ~25%, universal G3+ cytopenias post-Cy/Flu)

Counter-productive MoA: Moderate (Antigen-loss / HLA-LOH escape and post-lymphodepletion immunosuppression window; durability unproven, responders melanoma-skewed)

Overall: The highest-response on-axis option with a named sarcoma cohort, and the one with a fatality on the board — efficacy-first personas lead it, three dissent on the safety tail and n=28 imprecision.

Key references: PMID 40205198 · NCT03686124

2. brenetafusp (IMC-F106C, PRAME x CD3 ImmTAC) via NCT04262466 / successor IMC-P115C *(conditional on prame_hla positive)*

Likelihood of effect: Low-to-moderate: overall ORR 9-11%, benefit concentrated in a PRAME-positive subset (mPFS 4.5 vs 2.1 mo); melanoma-derived, cross-tumor.

Toxicity burden: Low (CRS predominantly G1-2, no G3+ CRS in cohort; rash ~70%, G3+ TRAE ~30%)

Counter-productive MoA: Low (On-target/off-tumor rash and antigen-density dependence; dissent is enrollability-flavored, no mechanism-level counter-productive vector)

Overall: The best-tolerated PRAME route with an off-the-shelf, no-lymphodepletion trade — but modest single-agent efficacy and a sarcoma enrollment slot that is currently closed.

Key references: doi:10.1200/JCO.2024.42.16_suppl.9507 · NCT04262466

3. IMC-P115C (next-gen PRAME x CD3 ImmTAC) via NCT07156136 *(conditional on prame_hla positive)*

Likelihood of effect: Unknown — first-in-human, no efficacy data; class bet on the ImmTAC platform and PRAME target, inferred from brenetafusp.

Toxicity burden: Low (no drug-specific data; class expectation predominantly G1-2 CRS and rash)

Counter-productive MoA: **Low** (Same on-target/off-tumor and antigen-density dependence as the ImmTAC class; no drug-specific counter-productive signal yet)

Overall: The recruiting-today PRAME ImmTAC when brenetafusp's sarcoma slot is closed — a platform-and-target bet with no efficacy readout of its own yet.

Key references: NCT07156136 · doi:10.1200/JCO.2024.42.16_suppl.9507

4. NW-101C (PRAME TCR-T) via NCT07266298 (pan-solid PRAME/HLA-A*02:01 basket) *(conditional on prame_hla positive)*

Likelihood of effect: Unknown — no NW-101C data exist; efficacy inferred entirely from the IMA203 PRAME TCR-T class precedent.

Toxicity burden: High (no drug-specific data; class expectation is the IMA203 TCR-T profile — CRS, ICANS, post-lymphodepletion cytopenias)

Counter-productive MoA: **Moderate** (Same antigen-loss / HLA-LOH escape and lymphodepletion immunosuppression risk as the PRAME TCR-T class; no drug-specific data)

Overall: A redundant PRAME TCR-T fallback if the IMA203 slot is inaccessible — but zero data under this name; the efficacy expectation is entirely borrowed.

Key references: NCT07266298

Evidence in detail

IMA203 (PRAME TCR-T)

Wermke et al. *Lancet Oncol* 2024

- **Disease context:** PRAME-positive HLA-A*02:01-positive refractory solid tumors (basket: melanoma, sarcomas including synovial sarcoma, ovarian, NSCLC, head and neck, uterine, others) (2L+) — Heavily pretreated PRAME+ HLA-A*02:01+ patients; subset of synovial sarcoma cohort enrolled; lymphodepleting Cy/Flu prior to infusion
- **n:** 28
- **Toxicities:** **CRS** (grade any): 100% (n=28); **CRS** (grade ≥3): 11% (n=28); **ICANS-like neurotoxicity** (grade any): 25% (n=28); **post-Cy/Flu cytopenia** (grade ≥3): 100% (n=28); **treatment-related death** (grade 5): 4% (n=28)
- **Efficacy:** **ORR** 54% (95% CI 34-73); **DoR** not reached months; **PFS** 5.1 months; **TRAE_rate**